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Consumer trust in B2C e-Commerce and the importance of social presence: experiments in e-Products and e-Services

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Abstract

Reducing social uncertainty—understanding, predicting, and controlling the behavior of other people—is a central motivating force of human behavior. When rules and customs are not sufficient, people rely on trust and familiarity as primary mechanisms to reduce social uncertainty. The relative paucity of regulations and customs on the Internet makes consumer familiarity and trust especially important in the case of e-Commerce. Yet the lack of an *interpersonal* exchange and the *one-time* nature of the typical business transaction on the Internet make this kind of consumer trust unique, because trust relates to *other people* and is nourished through *interactions* with them.

This study validates a four-dimensional scale of trust in the context of e-Products and revalidates it in the context of e-Services. The study then shows the influence of *social presence* on these dimensions of this trust, especially benevolence, and its ultimate contribution to online purchase intentions.
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1. Introduction

Research tells us that human beings like to reduce their social uncertainty. In other words, they seek ways to understand, predict, and occasionally attempt to control the behavior of other people. When social uncertainty cannot be reduced through rules and customs, people resort to trust and, to a lesser degree, to familiarity as major social complexity reduction methods [51]. Indeed, trust is among the most enduring characteristics of human interaction [4,51,70], especially when the expected outcomes of the interaction with others are not fully governed by rules and guarantees [4,42,43,79].

Trust is also a central element in many commercial activities [15,18,65], especially when the trusting party depends on, yet lacks control over the trusted party and where, consequently, trust that the trusted party will behave as expected can serve as a substitute for formal agreements in commercial exchanges [18,33,46,51].

Consistent with this observation, trust should be even more important in e-Commerce than in traditional commerce because of the paucity of rules and customs in regulating e-Commerce and because online services and products typically are not immediately verifiable. Moreover, online transactions lack the assurance provided in traditional settings through formal proceedings and receipts [20,65]. Indeed, research has shown that high levels of consumer trust encourage online purchase intentions [20,41] and help retain

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online customers [65], while the lack of it is the main reason individuals do not shop online [34]. There is also a growing recognition in industry circles that, as just argued, consumer trust facilitates e-Commerce (see US Better Business Bureau as reported in Cole [8]). Indeed, many dot.com failures have been attributed to the vendor's inability to create a strong trusting relationship with its customers [14], and, consistent with this observation, *only 5% of VISA International clients appear to trust e-Commerce*, a very low percentage compared with other financial transactions [49].

Yet, it must be recognized that the application of trust to e-Commerce does not apply perfectly to traditional business settings. Trust is an *interpersonal* determinant of behavior that deals with beliefs about the integrity, benevolence, ability, and predictability of other people [52,55]. However, in contrast to face-to-face commerce and to other applications of trust in the literature discussed below, there are typically no interpersonal interactions in e-Commerce, neither direct nor implied. Such interactions, or even cues relating to them, are notably missing from e-Commerce Websites [65].

How then does trust relate to e-Commerce where there is no interpersonal interaction and where there is no human agent toward whom this trust can be directed and on whose behavior it can be based? Given these unique circumstances, this study examines which of the established dimensions of trust in interpersonal relationships and in traditional Business-to-Consumer (B2C) commerce pertain to consumer trust in B2C Internet vendors (e-Trust). In doing so, this study extends the Familiarity and Trust Model of e-Commerce [20] in the following manner: (1) it embraces a multi-dimensional trust construct rather than a single dimension of trustworthiness to examine each dimension of trust on its own; and (2) it introduces a trust antecedent in the model which taps into the perception that, despite the lack of interpersonal interaction, there is still a sense of social presence on a Website.

These objectives were addressed through two free simulation experiments. The initial study built and verified a scale that captures the dimensions of trust in purchasing books on the Internet. The second study then replicated and extended the first study by adding social presence as an antecedent of trust in purchasing flight tickets online. The experiments utilized the Websites of two major players in this market-space, Amazon.com and Travelocity.com, respectively. Amazon.com sells tens of millions of books, CDs, and DVDs to more than 17 million customers, according to its Website. Travelocity.com sells airline tickets and associated services. *The Economist* recently ranked Amazon.com as the most visited US Website and Travelocity.com as the 10th [78]. There is also a growing recent interest in Travelocity.com within the airline industry [56,57] making the results of this study of particular interest to the industry.

The present study shows (1) that e-Trust is composed of four distinct beliefs dealing with the integrity, benevolence, ability, and predictability of the vendor; (2) that among these beliefs, integrity and predictability are the pertinent ones; (3) that the control variables disposition to trust and familiarity affect these beliefs; and (4) that e-Trust, especially the belief in benevolence, is increased by the perception of social presence in the Website.

2. Theoretical foundations and models

2.1. What is trust?

One of the central aspects of human behavior is the need to control and predict (or at least to understand) the social environment. Individuals have a need to "know" in advance how their behavior will influence the behavior of others, and how the behavior of others will consequently affect them. Rules and customs do provide some measure of social complexity reduction by regulating many aspects of social behavior, but properly understanding this social world is complex beyond human capacity because other individuals are essentially independent agents over whom one has little, if any, actual control and whose behavior is not always obvious or, indeed, rational. As a result, when interacting with others, individuals need to consider a vast number of possible interrelated behaviors that other individuals and organizations might exhibit [51].

Yet taking into consideration so many relevant interrelated behaviors is cognitively overwhelming. One way of coping with this social complexity is to presume that others will behave in a socially acceptable manner within the context of the relevant interaction and will not suddenly behave in unpredictable and socially unacceptable ways. When rules and customs are not enough, trust becomes the substitute guarantor that replaces relevant rules and customs as the guarantor that the expected outcomes of the relationship will materialize [42,43]. This assumption is the essence of trust [51].

Accordingly, trust is a context-dependent multidimensional social concept whose relevant significant dimensions depend on the circumstance of the interaction [6,19,50,69], and contains both *behavioral intentions* and *cognitive* elements [50,55,69].[2] The behavioral intentions aspects of trust deal with behavior that increases one's own vulnerability to others under conditions of interdependence; the cognitive aspects of trust deal with context-related beliefs about the trusted party that provide the context and justification for this behavior [50,69]. In general, research suggests

[2] Adding to this complexity is the lack of a standardized terminology. For example, the three trust beliefs of integrity, benevolence, and ability are called "trustworthiness" by Mayer et al. [52] and "antecedents of trust" by Jarvenpaa et al. [39], the latter using the term "trustworthiness" to reflect perceived behavior. Some earlier studies do not even differentiate between trust beliefs and trust. Some use the term "trust" in conjunction with "benevolence" (e.g., [48]) or "integrity" (e.g., [47,61,73], or as a mixture of ability and integrity (e.g., Ganesan [19]).

that the cognitive aspects of trust deal with beliefs that the trusted party will behave ethically [37] and will carry out expected commitments [4,13,48,52,71,73] under conditions of vulnerability and dependence [72]. This form of trust, discussed at length in the next section, deals typically with beliefs regarding the *ability, benevolence*, and *integrity* of the trusted party. Recent research also adds *predictability* to this list [52,55]. Although research has found many other situation-dependent beliefs, the first three beliefs seem to be common across many settings [30] and play a central role in management studies of trust where predictability is sometimes also added (e.g., [52,55]).

2.2. Definition of trust

The distinction between the behavioral and the cognitive aspects of trust began with Deutsch's [13] seminal study of trust in the Prisoners' Dilemma game [30]. Deutsch [13] concluded that trust is a set of *expectations* that lead to behavioral intentions that involve potential loss, because of the absence of control over those upon whom one depends. Based on an extensive literature review, Giffin [30] too defined trust as "reliance upon the characteristics of an object, or the occurrence of an event, or the behavior of a person in order to achieve a desired but uncertain objective in a risky situation" (p. 105). This trust, explains Giffin [30], typically consists of the three trust dimensions that were long ago set forth in Aristotle's *Rhetoric*: intelligence (corresponding to ability), good character (honesty and integrity), and good will (benevolence).

Examining trust in social exchanges, Blau [4] also concluded that trust contains three distinct beliefs: integrity, benevolence, and ability. Schurr and Ozanne [73] likewise define trust in this way. Rotter [71] also defined interpersonal trust as "an expectancy held by an individual or a group that the word, promise, verbal or written statement of another individual or group can be relied on" (p. 444). Again, trust is defined as a situation-specific expectation or belief that determines behavior. Luhmann [51], too, while not considering *ability,* made a distinction between the trust dimensions dealing with *integrity* and those dealing with *benevolence* in his analysis of the role of trust in social relations.

Later research has taken a similar view. Examining business activities, Zucker [86] has defined trust as a set of shared social expectations that are essential for and determine social behavior, enabling individuals to respond to each other without the explicit specification of contractual details. Similarly, Korsgaard et al. [45] argue that trust is the confidence that team members have in the good will and honesty (that is to say, benevolence and integrity) of their leader, while Hart and Saunders [33] have defined trust as the confidence that another party will behave as expected, combined with expectations of the other party's good will. This trust is composed of a perception of the partners' competence, openness (the willingness not to withhold information), caring (not taking unfair advantage), and reliability. Mishra [58], examining trust in the context of organizational response to crisis, applies the same dimensionality of trust. In a like fashion, McAllister [53] defines trust as "the extent to which a person is confident in, and willing to act on the basis of, the words, actions, and decisions of another" (p. 25), that is, as a behavioral intention based on beliefs. Kumar [46] and Ramaswami et al. [64] define trust as dependability based upon perceptions of procedural justice—fairness in managing a relationship, policies and procedures—and distributive justice—the perceived fairness of outcomes.

In a comprehensive review of management research on trust in a special issue on trust in the *Academy of Management Review*, Rousseau et al. [72] conclude that a common element in this research is that trust deals with "the intention to accept vulnerability based upon positive expectations of the intentions or behavior of another" (p. 395). Consistent with Lewis and Weigert [50], Rousseau et al. [72] also note that trust differs depending upon the history and nature of the relationship between the parties. In cases such as "initial trust" [55], trust is composed of (1) beliefs dealing with competence (ability), benevolence, honesty (integrity), and predictability, and (2) trusting-intentions; while in other cases, such as "opportunistic betrayal" [16], trust is composed of benevolence and integrity alone.

Aligned with these views is the marketing literature which develops two general approaches in its studies of trust: (1) as a behavioral intention and (2) as a set of beliefs [61]. In the latter school of thought, Schurr and Ozanne [73] define trust as the "belief that a party's word or promise is reliable and that a party will fulfill his/her obligations in an exchange relationship" (p. 940); this belief leads to behavioral intentions. Citing this definition, Dwyer et al. [15] expand on Blau's [4] work when they define trust as a set of beliefs relating to the exchange partner's ability and willingness to take part in the social exchange. Similarly, Anderson and Narus [2] argue that the essence of business exchanges is a firm's belief that another party will behave in a manner that will promote the well being of the firm and refrain from harmful actions. Moorman et al. [59,60] conceptualize trust in a similar manner. Trust is "a willingness to rely on an exchange partner in whom one has confidence" (p. 315) [59]. Morgan and Hunt [61] regard trust as confidence in another person's reliability and integrity. As in Moorman et al. [60], Ganesan [19] also defines trust as a willingness to depend upon another, based on beliefs or expectation resulting from the partner's experience, reliability and benevolence. This definition reflects two major dimensions of trust in the realm of marketing: credibility and benevolence. The former is composed of reliability and ability, a dimension Ganesan calls "objective credibility", whereas benevolence deals with motives.

Trust is a relatively new concept in IS research. Examining virtual teams in the Internet environment, Jarvenpaa et al. [39] found that trust (termed "antecedents of trust" by Jarvenpaa et al. [39]) consists of three distinct factors: ability, integrity and benevolence. These three beliefs affect

the members of virtual team's trusting behavioral intentions, meaning their willingness to depend upon other team members. In these virtual teams, integrity consistently had the strongest effect on trusting intentions while the importance of ability decreased over time. Gefen [20], in the context of e-Commerce, defines trust as a single dimension construct dealing with a consumer's assessment that the vendor is trustworthy, based on Luhmann's [51] definition of trust as a social complexity-reducing mechanism that leads to a willingness to depend on a vendor; this willingness is derived from the perception that the vendor will fulfill its commitments. Also based on Luhmann [51], a more recent study by Gefen [22] dealing with ERP adoption, conceptualized trust in a vendor's technical support team as composed of the three dimensions suggested by Giffin [30]—integrity, benevolence, and ability.

2.3. The varying dimensionality of trust

Table 1 summarizes the dimensions of trust as operationalized in previous research. Although there is an apparent agreement on the meaning of trust as an alternative to rules and customs, it should be noted that the dimensions that constitute it are context-specific [50,51]. To understand this varying dimensionality of trust one needs to go back to the psychological need behind trust.

2.4. Social presence

In the research cited above, however, the phenomenon studied was mostly an ongoing *interpersonal* interaction with another person or organization over extended periods of time. Indeed, trust is typically built gradually through extensive ongoing interactions that enable individuals to create reliable expectations of what other persons or organizations may do [4,51]. This is interesting from the perspective of the current study because the lack of an interpersonal interaction is a key defining characteristics of e-Commerce!

This *social context* is an important characteristic of trust [4,51]. This is also a common theme in the trust-building behaviors discussed in some of the articles in Table 1. The reason is that trust, in general, is built through constructive interactions with *other people* [4,18,51]. Since human interaction with the trusted party, whether face-to-face or by any other means, is a precondition of trust [4] the perception of a high degree of social presence, implying direct or indirect human contact, in the relationship should, arguably, contribute to the building of trust. Certainly, deliberately avoiding the creation of a social interaction and making the relationship devoid of a social presence, such as showing a "cold shoulder", reduces trust [4]. Extending this logic implies that another way in which trust in an e-Commerce Website may be built is through imbuing the medium of communication with a high social presence: the perception that there is personal, sociable, and sensitive human contact in the medium [74].

Social presence theory [74] describes how this social context affects medium use. Also related to media information richness theory [12,67,75], social presence theory argues that medium users assess the degree of social presence required by the task and fit it to the social presence of the medium, that is, how much a medium enables a communicator to experience communication partners as being psychologically present [74,81]. High social presence is typically found in face-to-face communication whereas low social presence is often found in e-mail (sans multi-media cues), and paper-based mail. Channel richness can vary depending on circumstances [77,85], however, and social presence can be high or low in any given Website.

Granted that an e-Commerce Website typically involves no actual interaction with other people, does not mean, however, that social presence cannot be embedded in a Website. If photos of people can convey a sense of personal, sociable and sensitive human contact, so too should multimedia Websites. Indeed, many Websites contain pictures of smiling people (a good example of this is http://virginmobileusa.com/) as they would be added in a typical brochure. This antecedent is of particular interest assuming that the perception of social presence can still be created despite the lack of actual human contact. Another way in which this is done in many Websites is through adding a "social touch" to the interaction, such as welcoming the consumer by name as he or she enters the Website and making Website and subsequent e-mail communications personalized. The two Websites used in this study adopted such a strategy, which resulted in variable responses to the social presence of the site.

Higher perceived social presence may also increase trust through its effect on increased electronic communication, as shown in e-mail interactions [28]. Communication is a necessary ingredient of constructive interaction. Indeed, research has found that richer media, that is, media higher in social presence, tend to be preferred in communications settings where the task is ambiguous and uncertain (see Straub and Karahanna [76], for a meta-analysis of this work).

Social presence should also build trust through the perception that the vendor is displaying through the Website a sense of personal, sociable, and sensitive human contact. From a consumer perspective, such attributes are clearly desirable. When dealing with consumer queries and grievances, they are even an expected aspect of any service or product. Hence, in accordance with Luhmann's [51] and Blau's [4] observation that trust is increased when the trusted party shows behavior or other indicators in accordance with one's expectations, the perception that the e-Vendor is embodying a high degree of social presence in the Website should increase consumer trust, to the degree that such indications are expected. This proposed, dynamic role of social presence adds a new perspective because of the generally and inherently lower levels of social presence in e-Commerce Websites.

Table 1
Trust dimensions in previous research

Source	Dimensions
Blau [4]	Integrity, benevolence, and ability
Giffin [30]	Integrity, benevolence, and ability
Rotter [71]	Predictability
Luhmann [51]	Integrity, and benevolence
Rotter [70]	Dependability
Schurr and Ozanne [73]	Predictability
Zucker [86]	Shared social expectations
Dwyer, Schurr and Oh [15]	Ability and willingness
Anderson and Narus [2]	Belief in trustworthiness
Crosby et al. [11]	Integrity and benevolence
Buttler [6]	Availability, competence (ability), discreteness, fairness, integrity, loyalty, openness, promise fulfillment, receptivity (all these factors are determinants of trustworthiness)
Moorman et al. [60]	Belief in trustworthiness
Moorman et al. [59]	Integrity and ability
Ganesan [19]	Experience (ability), reliability, and benevolence
Morgan and Hunt [61]	Reliability and integrity
Mayer et al. [52]	Ability, benevolence, and integrity (as antecedents of perceived trustworthiness)
Hosmer [37]	Expectation of "morally correct decisions" (p. 399)
Kumar et al. [48]	Honesty (integrity) and benevolence
Kumar et al. [47]	Honesty (integrity) and benevolence
Kumar [46]	Honesty and dependability
Korsgaard et al. [45]	Good will and honesty (i.e., integrity)
McAllister [53]	Trustworthy, caring
Mishra [58]	Competence (ability), openness, caring (benevolence), reliability
Ramaswami et al. [64]	Belief in trustworthiness
Hart and Saunders [33]	Predictability, competence, openness, caring, and good will
Elangovan and Shapiro [16]	Benevolence and integrity
Jarvenpaa et al. [39]	Ability, integrity and benevolence
McKnight et al. [55]	Competence (ability), benevolence, honesty (integrity), and predictability
Rousseau et al. [72]	Positive expectations
Gefen and Silver [27]	Ability, integrity, and benevolence
Jarvenpaa and Tractinsky [41]	Combination of trustworthiness, integrity and benevolence.
Kollock [44]	Trustworthiness
Gefen [20]	Belief that the e-Vendor is trustworthy
Ridings and Gefen [68]	Ability, integrity, and benevolence
Gefen [23]	Belief that the e-Vendor is trustworthy
McKnight et al. [54]	Competence, integrity, and benevolence
Gefen et al. [26]	Ability, integrity, and benevolence
Gefen et al. [25]	Belief that the e-Vendor is trustworthy
Pavlou [63]	Belief that the e-Vendor is trustworthy

2.5. Research model and hypotheses

Two experiments were conducted to test the research model. The objectives of study 1 were (1) to build a multi-dimensional scale of trust, (2) to verify the hypotheses regarding which trust dimensions affect purchase intentions in spite of the lack of actual human interaction, and (3) to confirm that trust, also as a multi-dimensional construct, is affected by the control variables familiarity and trusting disposition. The objective of study 2 was, first, to replicate study 1, and then to examine whether social presence has an effect on the verified scale of trust created in study 1.

Given the distinct possibility that a dishonest e-Vendor might have procedures that allow its employees not to keep promises to the customers, the belief in the integrity of the e-Vendor should be a central belief convincing the consumers that their expected outcomes from the interaction will be fulfilled, and should, therefore, be a significant dimension

of trust, in this case by reducing social uncertainty involved in breaking promises. Moreover, a dishonest e-Vendor may even make inappropriate use of credit card and personal information and could track purchase activity without prior approval. The reality of such threats is evidenced in Websites that automatically track e-Commerce shopping and then inappropriately use the information generated [62]. Trust in the integrity of the e-Vendor should decreases the social uncertainty involved in such behaviors, thus reducing the range of unacceptable social behaviors, and in doing so should assure the consumer of his or her expected outcomes when purchasing from the e-Vendor.

H$_1$. *Consumer trust in an e-Vendor's integrity increases intentions to purchase online from that e-Vendor.*

Like integrity, potential buyers must also believe in the predictability of the vendor. They need to expect that it will behave reliably, for instance, in delivering the goods and services on time by abiding to accepted and expected rules of conduct, and in doing so reduce social uncertainty relating to when and how these goods or services will be delivered. We postulate that, even in the online shopping environment, because there are human agents involved and the behavior of these agents can directly affect the consumers' outcome from the interaction, also predictability should be a significant characteristic of trust and increase consumer willingness to purchase online from the e-Vendor.

H$_2$. *Consumer trust in an e-Vendor's predictability increases intentions to purchase online from that e-Vendor.*

Also ability is hypothesized to have an effect on purchase intentions. The trust dimension of ability is the assessment that the trusted party knows its job and that this knowledge reduces the uncertainty that is involved in the trusting party obtaining its expected outcomes from the relationship by virtue of reducing the possible range of undesirable behaviors relating to inadequate ability that the trusted party may show. Since a lack of ability to perform the task properly should influence the trusting party's expected outcome in the case of e-Commerce, ability too should be a significant predictor of purchase intentions.

H$_3$. *Consumer trust in an e-Vendor's ability increases intentions to purchase online from that e-Vendor.*

The last trust dimension, benevolence, deals with the belief that the trusted party actually cares about the trusting party. Caring as an aspect of emphatic good service generally increases customer satisfaction and retention [23,83,84]. Benevolence reduces social uncertainty by allowing the trusting party to rule out undesirable behavior, namely the possibility that the trusted party will act with a short-term opportunistic profit motive while the trusting party is behaving with a long-term orientation. It too should accordingly be a significant predictor of purchase intentions.

H$_4$. *Consumer trust in an e-Vendor's benevolence increases intentions to purchase online from that e-Vendor.*

How then does trust arise in an e-Commerce environment? The literature suggests that trust in a typical business or social environment is created through two processes: constructive interaction [1,2,4,31,35,46,55,60,61] and interpersonal behaviors and cues that indicate trustworthiness [19,46,52,60,61,64]. These behaviors facilitate trust by confirming a belief in fair play—not being taken advantage of—and showing that the trusted party is investing its fair share in the relationship [19,43]. Another way trust can be built is through observed behavior that is in accordance with expectations, so that showing a trustworthy track record builds trust [4,27,51]. Many such behaviors and cues have been identified in the literature: (1) a trustworthy track record of previous behavior [46,53,86]; (2) the nature of the business interaction [19,27,33,39,46,64]; (3) showing a willingness to invest in a long-term relationship [2,4,13,35,46,61]; (4) investing beyond what was required by the initial contract [19]; (5) cooperation [11,15]; (6) staying in touch [11]; and (7) not demonstrating opportunistic behavior [61,82].

In accordance with Luhmann's [51] and Blau's [4] observation that trust is increased when the trusted party shows behavior or other indicators in accordance with one's expectations, the perception that the e-Vendor is embodying a high degree of social presence in the Website should also increase consumer trust, to the degree that such indications are expected. This proposed, dynamic role of social presence adds a new perspective because of the generally and inherently lower levels of social presence in e-Commerce Websites.

Conversely, when a Website conveys a *low social presence*, it is transmitting a "cold-shoulder" message. That message is not typically what builds trust [4]. Moreover, given that a personal, sociable service from a vendor is, in general, an integral part of the service consumers pay for, *consumers confronted with a site with low social presence* would have little reason to believe in the *integrity* of such an e-Vendor if the consumer believes that the e-Vendor is not servicing him or her in what has come to be expected. A consumer also has few reasons to believe in the *benevolence* of such a vendor to provide good service, given that the e-Vendor is not sensing and responding with good service. The *ability* of a vendor to provide service is also scrutinized by consumers when a Website does not appear to be responsive. Consumers may wonder whether they are dealing with an organization that has competent staff. And, finally, the consumer has little reason to believe in the vendor's *predictability*. Why should the consumer expect future predictable service given that current-day service standards are not being meet? Accordingly:

H$_5$. *Higher social presence embedded in a Website increases consumer trust in an e-Vendor's integrity.*

H$_6$. *Higher social presence embedded in a Website increases consumer trust in an e-Vendor's predictability.*

H$_7$. *Higher social presence embedded in a Website increases consumer trust in an e-Vendor's ability.*

H$_8$. *Higher social presence embedded in a Website increases consumer trust in an e-Vendor's benevolence.*

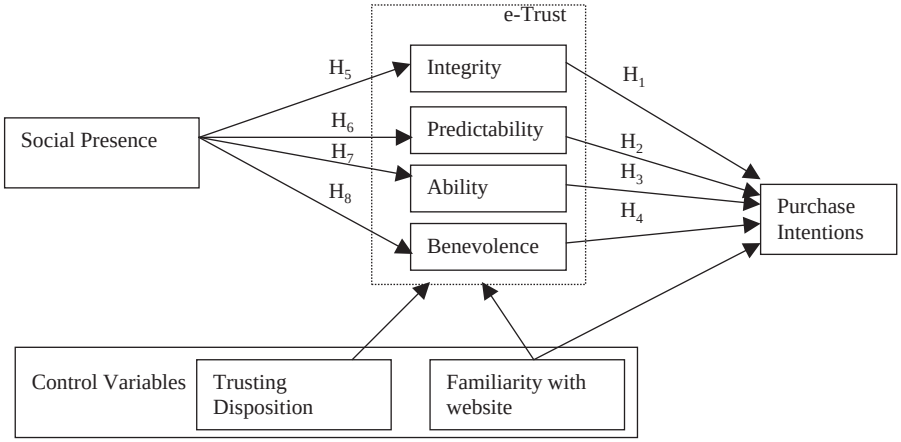


Fig. 1. Research model.

2.6. Control variables

The literature, specifically the Familiarity and Trust model [20] and its derivatives [25,26], suggest that trust also depends on a trusting disposition and familiarity, and that online purchase behavior also depends on familiarity. Accordingly, familiarity and trusting disposition are added as control variables based on [20]. A detailed discussion of the control variables appears in Appendix D. The research model is shown in Fig. 1. In Study 1 hypotheses 5–8 were not tested.

3. Methods and data analysis for study 1: Free simulation experiment using Internet book purchasing task

The research was conducted in two related, but distinct studies. In study 1, a four-dimensional scale of e-Trust was created. The reliability, convergent, and discriminant validity of the scales were then evaluated as well as whether the dimensions of trust are affected by the two e-Trust antecedents that were identified by previous research [20]. In study 1 hypotheses H_1 through H_4 were examined. Study 2 revalidated the four dimensions of e-Trust, examined these hypotheses, and extended the research model to include Social Presence as an additional antecedent of e-Trust with hypotheses H_5–H_8.

Both studies were free simulation experiments where the experimental setting duplicates or simulates, as closely as possible, a real world situation. The strength of this technique is that it allows the researcher to observe and measure subjects engaged in real world task [17]. Because no variables are specifically manipulated (only observed and measured), the technique is known as a "free" simulation experiment. The specific advantage of using a free simulation in the case of this research was to refresh, in the mind of the e-Consumers, the nature of the interaction with the e-Vendor, and, thus, increase the content validity of the study by collecting responses based on a recent activity. This is a true experiment in that manipulation of the subjects is occurring, although subjects are not formed into treatment groups, as in the most common lab experiments [17].

In selecting the sites to be included in this study, we deliberately chose e-Vendors that are well known and widely used. In fact, both were recently listed among the top 10 visited sites [78]. Selecting these sites controlled for extraneous variance dealing with e-Vendor size, risk in dealing with unknown vendors, and reputation [41].

3.1. Research design: study 1

3.1.1. Instrument and task creation

The purchase intentions scale, which dealt with consumer intentions to purchase books on a Website and to provide credit card information in the process, the familiarity scale, and the trusting disposition scale were adapted from the literature for the special circumstances of the selected Website [20]. Items for the four dimensions of trust reflect aspects of the dimensions as they apply to B2C e-Commerce. The items reflecting ability deal with the e-Vendor's knowledge, competence, and provision of good service. Items reflecting integrity deal with the e-Vendor's honesty and keeping of promises. The items reflecting benevolence deal with the e-Vendor's benevolence, willingness to assist and support, and with consideration toward the customer. The items reflecting predictability deal with knowing what to expect from the e-Vendor. The scales were created by selecting and adapting items from marketing research that dealt with service-providing or buyer–seller relationships [11,19,48,60,61] and that applied also to e-Commerce activity. It should be noted in this context that none of these marketing studies utilized the same scale. Moreover,

occasionally, even in the same study (e.g., [19]) there is no single scale for trust across scenario types. To these items additional items were added to reflect themes in the qualitative trust literature [4,18,36]. All items were statements and subject responses were offered on a 7-point scale ranging from Strongly Agree (1) to Strongly Disagree (7).

The items were pilot tested with two experienced web shoppers to ensure linguistic and thematic clarity, and were then pre-tested with 95 subjects from the population (MBA students) in an experiment identical to the one that was later conducted to collect the actual data (see details below). The factorial validity of the proposed scale was examined with a Principal Components factor analysis, and two items, IN4 and AB2, that loaded highly on more than one factor were dropped because they did not load cleanly in the factor analysis. The scales are shown Appendix A.

3.1.2. Experimental procedure: study 1

All told, 250 MBA subjects in a business school in the Mid-Atlantic region of the USA took part in the first experiment. The classes in which the experiments were performed did not deal with e-Commerce and the subjects were not rewarded for their participation. There was no overlap between subjects who took part in the pre-testing and those who took part in the first main data collection. The subjects were given a brief introduction telling them that the study dealt with e-Commerce and asking for their assistance. The experimenter then demonstrated the features of www.amazon.com, and requested subjects to navigate to the Website on their in-lab PCs and go through the procedure of buying the textbook of the course they were currently taking, but not to actually purchase the book. This ensured that the subjects' memory was refreshed about the Website and its procedures prior to completing the experimental instrument.

At the conclusion of the task, subjects were asked to fill in the experimental instrument. Only after filling it and returning the instrument were the subjects debriefed about the research model and the hypothesized role of trust. In this manner, prior knowledge could not affect the responses. In all, 90% of the subjects completed the task and gave complete enough data for inclusion in the study resulting in 226 usable questionnaires. The subjects were mostly in their late twenties ($N = 95$), early thirties ($N = 71$), or late thirties ($N = 28$). Of those who disclosed their gender, half were men ($N = 101$) and half were women ($N = 100$). 130 subjects had purchased at least once previously at www.amazon.com.

3.1.3. Data analysis: study 1

The research model was examined using PLS. PLS combines a factor analysis with multiple linear regressions to estimate the parameters of the measurement model (item loadings on constructs) together with those of the structural model (regression paths among the constructs) by minimizing residual variance. The t-values are then estimated using a jackknife method [7,80]. PLS is especially suited for exploratory research [7,24,29,80], such as the current study.

In addition to parameter estimation, PLS can also test the convergent and discriminant validity of the scales. This is shown when: (1) measurement items load more strongly on their assigned construct than on the other constructs in a Confirmatory Factor Analysis (CFA), and when (2) the square root of the Average Variance Extracted (AVE) of each construct is larger than its correlations with the other constructs [7,29]. Appendix B shows the results of this PLS CFA: all the measurement items loaded considerably more strongly on their respective factor than on the other constructs. All the items loaded very significantly, with p-values well within the 0.01 level, on their assigned construct. Table 2, later, shows the inter-construct correlations and square root of the AVE (in the diagonal grayed boxes).

Table 2 shows that all constructs share considerably more variance with their measurement items than with the other constructs. Table 2 also shows that all the PLS composite construct reliability coefficients are in the upper 0.80s and 0.90s. The descriptive statistics show that the subjects had a positive trusting disposition, were familiar with Amazon.com, showed proclivities toward trusting in all four dimensions of trust, and reported that they were likely to purchase at the Website.

The research model with the PLS jackknife estimates is presented in Table 3. Coefficients supported at a 0.01 level are shown with a double asterisk. The analysis shows that Purchase Intentions was, as hypothesized, affected by Familiarity and by e-Trust ($R^2 = 37\%$). Among the dimensions of e-Trust, only Integrity and by Predictability affected Purchase Intentions. Most of the dimensions of e-Trust were affected, albeit not strongly, by Trusting disposition and by Familiarity.[3] The degrees of explained variance of the dimensions of trust were: Predictability 13%, Benevolence 16%, Integrity 16%, and Ability 14%. There are few surprises in these tests as they reconfirm linkages in Gefen's [20] original *OMEGA* study, albeit extending it considerably by examining trust as a four-dimensional construct, rather than as a single dimension.

[3] In exploratory, post hoc tests, additional analyses were also performed on the data separating between experienced and inexperienced consumers based on whether the consumer reported previous purchases at the e-Vendor. These additional analyses used stepwise linear regressions that replicated the linkages in the original research model. The data analyses of both experienced and inexperienced users showed equivalent significant linkages to Table 4, the only major exceptions being that: (1) among experienced consumers ($N = 130$), benevolence *reduced* purchase intentions, and (2) among the inexperienced subjects ($N = 96$) predictability was an insignificant predictor of purchase intentions.

Table 2
Composite PLS reliability and inter-construct correlations and square root of AVE

Amazon	Intended purchase	Predict-ability	Bene-volence	Integrity	Ability	Trusting disposition	Familiarity
Intended purchase	0.906						
Predictability	0.402	0.906					
Benevolence	0.297	0.529	0.830				
Integrity	0.482	0.587	0.619	0.899			
Ability	0.350	0.513	0.609	0.568	0.861		
Trusting disposition	0.191	0.277	0.333	0.269	0.208	0.829	
Familiarity	0.476	0.256	0.251	0.322	0.329	0.150	0.868
PLS reliability[a]	*0.901*	*0.902*	*0.899*	*0.926*	*0.895*	*0.907*	*0.924*
Mean (std.)	2.83 (1.36)	3.09 (1.09)	3.14 (1.00)	2.97 (0.95)	2.65 (0.94)	3.37 (1.14)	3.00 (1.53)

[a]PLS reliability is calculated as: $\rho = (\Sigma\lambda_i)^2/((\Sigma\lambda_i)^2 + \Sigma\varepsilon_i)$.

Table 3
Study 1 (Amazon.com) findings: path coefficients

Dependent	Independent					
	IN	PRED	AB	BEN	Trusting disposition	Familiarity
Purchase Intentions (USE)	0.31**	0.16**	n.s.	n.s.		0.35**
Integrity (IN)					n.s.	0.29**
Predictability (PRED)					0.24**	0.22**
Ability (AB)					0.16**	0.30**
Benevolence (BEN)					0.30**	n.s.

n.s. = insignificant.
**Significant at the 0.01 level.

4. Methods and data analysis for study 2: free simulation experiment using internet flight purchasing task

4.1. Research design: study 2

Having validated the e-Trust scales in the *e-Product space*, the objectives of the second study were (1) to replicate study 1, to extend the generalizability of the proposed trust scales to equivalent activities in the *e-Services space*, and (2) to extend the testing of the model by examining whether social presence affects e-Trust. As noted above, the low social presence in a typical Website as compared with a typical brick-and-mortar storefront selling the same services or products is one of the major differentiating characteristics of the Web-based "new- economy".

4.2. Instrument adaptation: study 2

Study 2 used exactly the same scale item wording as study 1, other than having "Travelocity" replace "Amazon" and "flight tickets" replace "books." The experimental instrument also contained scale items measuring Social Presence. The Social Presence scale presented in Table 4 was adapted from Gefen and Straub [28].

Table 4
Additional scale used in study 2

Social presence	Code
There is a sense of human contact in the Website	SP1
There is a sense of personalness in the Website	SP2
There is a sense of sociability in the Website	SP3
There is a sense of human warmth in the Website	SP4
There is a sense of human sensitivity in the Website	SP5

4.3. Experimental procedure: study 2

The second experiment was conducted with 171 other MBA subjects in the same business school in the Mid-Atlantic region of the USA, using the same Internet-accessible computer lab. These classes, likewise, did not deal specifically with e-Commerce and the students were not rewarded for their participation. As in study 1,

Table 5
Composite construct reliability and inter-construct correlations and square root of AVE

Travelocity	Intended purchase	Predict-ability	Benevol-ence	Integrity	Ability	Trusting disp.	Familiarity	Social presence
Intended purchase	0.911							
Predictability	0.423	0.902						
Benevolence	0.323	0.446	0.833					
Integrity	0.455	0.502	0.591	0.805				
Ability	0.392	0.580	0.453	0.633	0.819			
Trusting disp.	0.349	0.303	0.313	0.357	0.406	0.824		
Familiarity	0.372	0.171	0.164	0.159	0.097	0.058	0.856	
Social presence	0.329	0.294	0.481	0.308	257	0.288	0.145	0.836
Construct reliability	*0.907*	*0.897*	*0.901*	*0.846*	*0.859*	*0.894*	*0.916*	*0.919*
Mean (std.)	3.25 (1.52)	3.34 (1.20)	3.39 (1.33)	3.04 (1.02)	2.79 (1.01)	3.45 (1.15)	3.29 (1.66)	4.49 (1.56)

the subjects were given a brief introduction telling them that the study dealt with e-Commerce and asking their assistance. The experimenter then demonstrated the use of www.travelocity.com to check for flights and then book, albeit not to purchase, a round-trip flight from NYC to London UK in two weeks' time. The subjects were then requested to navigate to the Website and go through the procedure of buying an equivalent ticket, but not actually purchasing it. Having completed the task, the subjects were asked to complete the experimental instrument. With a 93% response rate, the sampling resulted in 161 usable instruments. Subjects were mostly in their late twenties ($N = 56$), early 30s ($N = 56$), or late thirties ($N = 21$). Of those who disclosed their gender, 61% were men ($N = 89$) and 39% were women ($N = 57$).

4.4. Data analysis: study 2

As in study 1, data were examined using PLS. The CFA (see Appendix C) shows that all the measurement items load much more strongly on their respective construct than on the other constructs. All the items loaded very significantly on their respective assigned construct, with p-values well within the 0.01 level. Table 5, below, shows the inter-construct correlations and square root of the AVE (in shaded boxes in the diagonal). Table 5 shows that all constructs share considerably more variance with their measurement items than with the other constructs, showing convergent and discriminant validity. The PLS composite construct reliability coefficients are in the 0.80s and 0.90s. The descriptive statistics show that the subjects had an equivalent positive trusting disposition as subjects in study 1; were familiar with the vendor; were slightly less trusting in most dimensions of trust; and reported that they were very likely to purchase at the Website although they assessed that its overall social presence as low.

The research model with the PLS jackknife estimates is presented in Table 6. The analysis shows that Purchase Intentions was affected by Integrity and Predictability and by Familiarity ($R^2 = 34\%$). The dimensions of e-Trust, except for Benevolence, were affected by Trusting disposition. Social Presence affected Integrity, Benevolence, and Predictability, but not Ability. Only Predictability, however, was affected by Familiarity. [4] Explained variance of the trust dimensions were: Predictability 15%, Benevolence 27%, Integrity 19%, and Ability 19%.

5. Discussion, limitations, and implications

Academic research (e.g., [20,41,65]) and industry reports (e.g., BBC [3], Cole [8]) claim that trust is important in e-Commerce. This study upholds this thesis, showing that 34–37% of the variance of consumer intentions to purchase an e-Product in study 1 and an e-Service in study 2 were explained by elements of e-Trust and the control variable Familiarity. Data analyses of two experiments show that our research model for e-Trust in the B2C space generally holds.

The study contributes to explanatory models of e-Trust by refining the models offered by previous research, by expanding the theory-base to include dimensions of trust, and by adding the dimension of social presence as another antecedent of e-Trust. The importance of each e-Trust dimension is explained in the context of reducing the social

[4] As in study 1, the data were re-analyzed splitting the data into experienced ($N = 104$) versus inexperienced ($N = 57$) consumers, according to whether they self-reported previous purchases at the specific e-Vendor. The data were analyzed with stepwise regression, although special caution needs to be exercised given the sample sizes. These analyses show equivalent effects to the model above, except that (1) here too, benevolence among experienced consumers had a *negative* effect on purchase intentions, and (2) among experienced consumers, predictability was not a significant predictor of purchase intentions, perhaps because of the type of service Travelocity.com sells and the fact that variance in the quality of bookings are co-dependent on partners.

Table 6
Study 2 (Travelocity.com) findings: path coefficients

Dependent	Independent						
	IN	PRED	AB	BEN	Trusting disposition	Social presence	Familiarity
Purchase intentions (USE)	0.26**	0.19**	n.s.	n.s.			0.29**
Integrity (IN)					0.24**	0.21**	n.s.
Predictability (PRED)					0.24**	0.21**	0.13**
Ability (AB)					0.36**	n.s.	n.s.
Benevolence (BEN)					n.s.	0.41**	n.s.

n.s. = insignificant.
**Significant at the 0.01 level.

uncertainty involved in achieving the consumers' outcomes from the interaction in the unique context of e-Commerce with its lack of an interpersonal contact.

The studies also reinforced the interpretation that e-Trust is a multi-dimensional construct, but showed that only its dimensions of integrity and predictability are significant antecedents of purchase intentions in e-Commerce. As explained above, trust is a set of beliefs about *other people* that serve to justify an interaction with these persons, based on the belief that they will behave socially as expected and in doing so will provide the trusting party with its expected outcomes. Arguably, depending on the trusting party's expectations of the trusted party and how these expectations relate to the trusting party's hoped for outcomes, some trust beliefs should be more important than others in each particular context. The more important beliefs among the four trust beliefs may be those on which the trusting party depends on the trusted person or persons to obtain its expected outcomes from the interaction.

High correlations between the dimensions of trust may account for lack of significance of the ability and benevolence dimensions. For this reason, it is not possible to make too much of the lack of significance here even though the power is reasonable, given the sample size. In situations with high multicollinearity, least-squares techniques cannot hold variables constant, and this could lead to artifactual results.

With respect to the control variables, the data analyses show, that, in accordance with the literature, e-Trust in both studies was affected by a general disposition to trust [55,71] and to a lesser extent by familiarity [51]. The importance of a trusting disposition is not surprising given that there is little to no personal interaction with the two e-Vendors, which forces the customers to rely on alternative, non-interpersonal interaction based, antecedents when deciding on their degree of trust in the e-Vendor. The addition of social presence as another antecedent of e-Trust in the context of e-Commerce extends previous research in showing that although a Website is typically devoid of actual human interaction, nonetheless, *the perception that there is a social presence does in itself increase e-Trust*. This tentative conclusion means that the perception of social presence in a Website, the possible semblance of an interpersonal interaction, is probably important in e-Commerce even though consumers typically interact with a Website rather than with another person.

When the data were analyzed post hoc to differentiate between experienced and new consumers, results revealed that in both datasets benevolence decreased purchase intentions among experienced consumers (although caution is called for given the lower sample sizes in these post-hoc analyses). In retrospect these results are not so surprising. Integrity and predictability are among the expected characteristics of the way a typical vendor treats all customers, including new ones. On the other hand, showing benevolence is not. As when people interact in general, indications of benevolence when caring is not called for are often interpreted as attempts to establish superior social standing, not as showing caring. As such, these cues inhibit people's willingness to take part in an interaction with that person [4]. Consequently, it is not surprising that experienced consumers' behavioral intentions are in accordance with what would be expected of typical wary consumers. After all, even in the bricks-and-mortar world, when a salesperson shows exaggerated caring, it is treated with suspicion and is thus likely to turn customers away.

Taken together, the two studies also provide an initial verified scale of e-Trust that shows equivalent convergent and discriminant validity and reliability with regard to both service and product e-Vendors. The fact that the instrument was valid across settings, times, and persons [9] means that it might be generalizable to other B2C e-Commerce situations, although, clearly, additional research is needed in this regard. There are many instruments in the literature that develop some of the components of trust, but the present e-Trust instrument is among the most comprehensive of these attempts, to date.

5.1. Limitations and future research

Several limitations and issues need to be mentioned and discussed. The use of graduate business school students as

surrogates for e-Consumers might raise issues of external validity [32] unless the students do, in fact, represent the target population [38], which, arguably, they might in this case [20]. Although there is no reason to believe that the average profile of the MBA students does not represent typical purchasers of online products and services of the types examined, as in other cases [66], additional research should examine whether these results apply equally well to other consumers in e-Commerce situations. Related to this limitation is the possible sample bias, the selection of a sample of experienced buyers. It is quite possible that both familiarity and the trust dimensions work differently with inexperienced buyers. Additionally, even though the pattern of results was consistent in both studies, the possibility of insignificant paths may be due to statistical issues. Additional research is needed here too.

The emphasis of this study was on assessing the impact of trust beliefs on e-Commerce. The data, accordingly, examined trust and related constructs. There are, however, many other issues affecting e-Commerce aside from consumer trust, such as consumer loyalty [65], product perceptions about the available variety, price, and quality, as well as the compatibility, playfulness and effort in the shopping experience, combined with customer service and perceived risk [10,40]. These variables are clearly important in e-Commerce. However, given that the focus of the study was on the unique aspects of trust in an e-Commerce setting (unique because there is no interpersonal interaction), these aspects were not studied. Hence, although this study reports relatively high R^2, expanding the research model to include these additional issues, which being significant in commerce in general might apply also to e-Commerce, could provide an invaluable tool to both industry and research.

There are also many contingencies that will undoubtedly affect the level of trust that e-Consumers have in e-Vendors. A short list would include: branding, the type of the item being transacted, the size of the purchase, previous history with e-Vendor, the role of trusted third parties, the use of new reputation mechanisms, the professional appearance of the Website, the speed of loading for the site, articles written about the company or its Website, word of mouth references or discussions of experiences or beliefs, the frequency, nature, and placement of ads for the site. This study did not attempt to address all these possible antecedents of trust but to examine the dimensions of trust itself in this unique e-Commerce environment, where there is a lack interpersonal interaction but where nonetheless trust as a set of interpersonal beliefs still applies. Additional research is needed to sort out the effects of such conditions on trust-building and the trust that users develop in e-Vendors.

Of course, there is a mirror image to issues surrounding consumer trust. Vendors also have to trust, both in the integrity of the transaction and in the ability of the consumer to pay. Credit checks may help in the latter case, and encryption and third party guarantors such as certificate authorities in the former, but there is still a possibility of fraudulent transactions. Cyberspace does not now permit the use of photo IDs of the consumer, as in physical retail establishments, and hackers can easily masquerade as legitimate users in the absence of digital signatures. This issue cuts both ways as consumers can also be duped by phony Websites. In short, there are many issues that need to be explored in the domain of e-Trust.

The potential importance of social presence raises the question of how a Website can be designed to increase social presence and what additional features, such as a toll-free number, it should include in order to increase its social presence. Additional research is needed to investigate these aspects.

5.2. Implications for researchers

Previous research examined trust as a single-dimension construct (e.g., [20,41,65]) dealing primarily with risk. This study takes a different view, showing that e-Trust is multi-dimensional and that its dimensions should be significant to the extent that they help reassure the customers about expected outcomes. Accordingly, the meaning and consequences of trust are better understood when each dimension is viewed separately. The potential value of this scale is increased by its robust psychometric properties over two different cases. Amazon.com sells a commodity product (mainly books), while Travelocity.com sells services (mainly airline reservations).

In the two independent forms of e-Commerce (e-Products and e-Services) where trust was examined, it consisted of the same dimensions of ability, integrity, benevolence, and predictability. The two studies also show, consistently, that integrity and predictability, together with familiarity, were significant antecedents of purchase intentions. And, in both studies, the belief in the e-Vendor's benevolence among experienced consumers reduced their purchase intentions. This suggests that although trust is important, as previous research found and as industry reports state, it is not a single monolith construct with a single effect. Rather, as revealed in this study, trust is multi-dimensional in both its measurement and its structural effects, and the significance of the effect of each dimension of trust depends on its specific applicability of the human-related expectation it represents.

Another contribution is the addition of social presence as perceived in the Website by the e-Consumer in study 2. Additionally, some trust antecedents have been identified and examined as control variables, namely, trusting disposition and familiarity. In this regard, the study justifies Morgan and Hunt's [61] claim that trust (as a set of beliefs) is a key mediating variable in relationship models and that overlooking the role of trust "results in flawed conclusions" (p. 31). Toward this end, this study has proposed and demonstrated a model of its effects.

On a broader level, the study shows that trust is a multi-dimensional construct but that—in accordance with the theory-base on trust being important in interpersonal

relationships because it reduces relevant aspects of social uncertainty [51]—only those dimensions of trust that are pertinent to the expected outcome of the interaction affect people's intended behavior.

The exploratory results regarding the effect of benevolence on experienced consumers is also interesting within the theory-base. As Blau [4] suggested, trust affects behavioral intentions when the trusted party behaves or indicates that it will behave in accordance with what is expected of it within the context of relevant behaviors and cues. But this does not mean that behaving in what would otherwise be generally considered a trustworthy manner, for instance by being benevolent, will necessarily encourage other people to take part in the transaction. Such behavior and cues only encourage behavior to the extent they are expected from the trusted party. And, if these exploratory results can be generalized, behaving or showing cues in such a manner may actually inhibit people from taking part in a transaction when they are not expected of the trusted party, because showing caring when it is not expected might be interpreted as indicating superior social standing and power rather than reducing social uncertainty [4].

5.3. Implications for practitioners

Although additional research is needed before definitive implications can be stated, the latter conclusion is interesting from a practitioner point of view too. One might be inclined to believe based on the literature that behavior and cues that show ability, integrity, predictability, and benevolence should be encouraged among e-Vendors because trust is a necessary precursor of online purchase [65], and may be, at least according to Fukuyama [18], the single most important factor directly affecting organizational and national prosperity and adaptability, including the acceptance of new IT.

However, to the extent that the exploratory results can be generalized, this principle should be qualified to only those cues and behaviors that are expected of the human side of the e-Vendor, probably because people are naturally suspicious of unmitigated and unexpected trustworthy cues and behaviors. Indeed, theory [4] suggests that showing excessive caring can actually inhibit people's intentions to participate in transactions because it is might be translated as a show of superior social standing. Nonetheless, practitioners should also be aware that in other studies dealing with trustworthy behavior, the data suggested that it is the *interpretation* of the motive behind the observed cues and behaviors that might determine what people do, rather than the perceived cues or behaviors themselves [21].

As more and more e-Vendors are realizing the importance of promoting trust online [5], this study provides some guidelines on how to send such a message. e-Trust appears to result, among other factors, from familiarity, and from cues and perceptions of social presence. Does this mean that new e-Vendors should apply interfaces and procedures that consumers are already used to from other e-Vendors? Does this imply that e-Vendors should try and maintain the same interface and work procedures over time? The data tentatively seem to suggest so. The finding about social presence also might have far reaching implications for Internet-based vendors because of the noted lack of social presence in Websites. Given these tentative conclusions, e-Commerce vendors might wish to invest in creating and maintaining effective social presence channels with potential consumers.

6. Conclusion

Trust, according to Fukuyama [18], is the lubricant of trade and, in this way, determines the wealth of nations. The present study lends support to this assertion, but also qualifies it by showing that e-Trust is not a single monolith belief but a set of beliefs of varying importance depending on their relevance to the expected outcome. The study also hints at how e-Trust can be increased. Both findings are crucial in articulating a basic understanding of e-Trust for e-Commerce managers as well as for academics investigating this phenomenon.

Appendix A. Items used in Study 1

(Study 2 substituted "Travelocity" for "Amazon" and "flight tickets" for "books".)

Scales and items	Code
Integrity	
Promises made by Amazon.com are likely to be reliable	IN1
I do not doubt the honesty of Amazon.com	IN2
I expect that Amazon.com will keep promises they make	IN3
I expect that the advice given by Amazon.com is their best judgment	IN4 (dropped)
Benevolence	
I expect I can count on Amazon.com to consider how its actions affect me	BEN1
I expect that Amazon.com intentions are benevolent	BEN2
I expect that Amazon.com puts customers' interests before their own	BEN3
I expect that Amazon.com is well meaning	BEN4

Ability	
Amazon.com is competent	AB1
Amazon.com understands the market it works in	AB2 (dropped)
Amazon.com knows about books	AB3
Amazon.com knows how to provide excellent service	AB4
Predictability	
I am quite certain about what Amazon.com will do	PRED1
I am quite certain what to expect from Amazon.com	PRED2
Intended purchase	
I am very likely to buy books from Amazon.com	USE1
I would use my credit card to purchase from Amazon.com	USE2
Trusting disposition	
I generally trust other people	DIS1
I tend to count upon other people	DIS2
I generally have faith in humanity	DIS3
I feel that people are generally well meaning	DIS4[a]
I feel that people are generally trustworthy	DIS5[a]
I feel that people are generally reliable	DIS6
Familiarity	
I am familiar with searching for books on the Internet	FAM1
I am familiar with buying books on the Internet	FAM2
I am familiar with Amazon.com	FAM3
I am familiar with inquiring about book ratings at Amazon.com	FAM4

[a] Items DIS4 and DIS5 were dropped so the trusting disposition scale would match Gefen [20].

Appendix B. PLS confirmatory factor analysis of Amazon.com data

	Ability	Benevolence	Familiarity	Integrity	Intended purchase	Predictability	Trusting disposition
AB1	**0.8574**	0.5323	0.3409	0.5166	0.3084	0.4654	0.1891
AB3	**0.8310**	0.4395	0.1698	0.4242	0.2529	0.3463	0.1990
AB4	**0.8923**	0.5782	0.3086	0.5120	0.3321	0.4900	0.2013
BEN1	0.5106	**0.8145**	0.2239	0.5455	0.2868	0.4547	0.2678
BEN2	0.5350	**0.8388**	0.2391	0.5072	0.3004	0.3945	0.2725
BEN3	0.6054	**0.8311**	0.1849	0.5323	0.1904	0.4918	0.2508
BEN4	0.3731	**0.8358**	0.1753	0.4715	0.1881	0.4240	0.3306
FAM1	0.3130	0.2317	**0.9041**	0.2930	0.4318	0.2120	0.1142
FAM2	0.2456	0.1861	**0.8900**	0.2923	0.5023	0.2481	0.1299
FAM3	0.2666	0.1814	**0.8251**	0.1963	0.3145	0.1399	0.0713
FAM4	0.3144	0.2635	**0.8489**	0.3170	0.3792	0.2676	0.1912
IN1	0.5014	0.5187	0.3418	**0.8843**	0.4339	0.4947	0.1847
IN2	0.4952	0.5176	0.2589	**0.9115**	0.4449	0.5685	0.2620
IN3	0.5348	0.6335	0.2675	**0.8998**	0.4207	0.5186	0.3044
USE2	0.3026	0.2644	0.4285	0.4352	**0.9030**	0.3447	0.2144
USE1	0.3317	0.2724	0.4327	0.4378	**0.9082**	0.3836	0.1568
PRED3	0.4807	0.5215	0.2285	0.5608	0.3174	**0.8925**	0.2671
PRED4	0.4524	0.4427	0.2350	0.5075	0.4065	**0.9199**	0.2658
DIS1	0.1409	0.2439	0.1306	0.1929	0.1322	0.2012	**0.8401**
DIS2	0.1914	0.2665	0.1173	0.1878	0.1970	0.2576	**0.8265**
DIS3	0.1955	0.3130	0.1322	0.2901	0.1490	0.2583	**0.8646**
DIS6	0.2152	0.2862	0.1175	0.2368	0.1960	0.2476	**0.7820**

Appendix C. PLS confirmatory factor analysis of Travelocity.com data

Travelocity	Ability	Benev-olence	Famili-arity	Integrity	Intended purchase	Predict-ability	Trusting disposition	Social presence
AB1	**0.8516**	0.4918	0.1043	0.5787	0.3268	0.5368	0.4205	0.2821
AB3	**0.8266**	0.2456	0.0084	0.5745	0.3433	0.4008	0.3276	0.1078
AB4	**0.7781**	0.3536	0.1260	0.3798	0.2922	0.4816	0.2209	0.2321
BEN1	0.3398	**0.8241**	0.1854	0.5199	0.3598	0.3971	0.3158	0.3753
BEN2	0.3946	**0.8356**	0.0659	0.4440	0.2562	0.3394	0.2226	0.4424
BEN3	0.4782	**0.8468**	0.1700	0.5342	0.2389	0.4060	0.2925	0.3978
BEN4	0.2918	**0.8264**	0.1193	0.4655	0.2034	0.3360	0.1967	0.3875
FAM1	0.1458	0.1618	**0.8794**	0.1616	0.2692	0.1180	0.0072	0.0944
FAM2	0.0304	0.1700	**0.9013**	0.1469	0.4186	0.1750	0.0525	0.1470
FAM3	0.0536	0.0670	**0.7351**	0.0957	0.3167	0.1891	0.1251	0.1816
FAM4	0.1208	0.1560	**0.8964**	0.1341	0.2246	0.0847	0.0068	0.0574
IN1	0.5190	0.4214	0.0474	**0.8012**	0.2460	0.4459	0.3109	0.2571
IN2	0.5470	0.5394	0.2327	**0.8704**	0.4762	0.4052	0.3195	0.2791
IN3	0.4625	0.4508	0.0553	**0.7384**	0.3326	0.3735	0.2253	0.2010
USE1	0.3595	0.2412	0.3173	0.4296	**0.9149**	0.4258	0.2881	0.2707
USE2	0.3547	0.3496	0.3615	0.3984	**0.9077**	0.3428	0.3490	0.3303
PRED1	0.4988	0.3819	0.0439	0.4216	0.3420	**0.8810**	0.2675	0.2617
PRED2	0.5448	0.4200	0.2442	0.4792	0.4144	**0.9223**	0.2783	0.2688
DIS1	0.3447	0.2271	0.0573	0.3225	0.3185	0.2845	**0.8502**	0.1918
DIS2	0.2527	0.2268	0.0663	0.1665	0.1855	0.2350	**0.6951**	0.2125
DIS3	0.3297	0.2340	0.0677	0.2861	0.3373	0.2034	**0.8479**	0.2992
DIS6	0.3916	0.3297	0.0125	0.3656	0.2955	0.2711	**0.8895**	0.2524
SP1	0.1724	0.4266	0.1468	0.2171	0.2919	0.2042	0.1983	**0.8761**
SP2	0.2729	0.3530	0.1388	0.2849	0.3214	0.2263	0.1986	**0.8844**
SP3	0.2309	0.4793	0.1154	0.3004	0.2720	0.2761	0.2378	**0.8892**
SP4	0.1155	0.1769	0.1054	0.0426	0.1911	0.1781	0.2412	**0.5685**
SP5	0.2472	0.4777	0.1128	0.3348	0.2906	0.3178	0.3389	**0.9107**

Appendix D. Control variables

D.1. Trusting disposition

Despite the importance of long standing social interactions, the literature suggests that trust depends in part also on a trusting disposition. Depending on the their cultural background [4], their faith in humanity and their socialized trusting stance [56,70], individuals typically enter into relationships with a certain degree of initial trust [52,56,70]. This initial trust is not based on the specific trusted person or organization; rather, it is based on the individual's lifelong socialization and experiences that lead him/her to believe to some degree or another that people in general can be trusted. A trusting disposition is one way to overcome the "Catch 22" problem as it relates to interpersonal trust. Without trust, most interpersonal interactions cannot occur, but without these interpersonal interactions the trust that is required for these interactions cannot be built. Reliance on a trusting disposition solves this "Catch 22" problem. It is a willingness to give a "trust credit" without justifying it on prior experience with the particular would-be trusted party. This trust is based on a trusting disposition that is the consequence of the lifelong experience of the trusting party; it is applicable, therefore, even when there is no prior experience with the specific would-be trusted party [56,70].

D.2. Familiarity

This construct is also an antecedent of trust because it enables people to place their trust beliefs about the future into a context which clarifies the specifics of what they expect of others: the what, when, where, and how [52]. This view is also in accordance with Blau [4] who explains that trust is built through successful previous interactions with the trusted party, the subsequent acquaintance with the trusted party and the knowledge gained about its trustworthiness and what to expect of it.

Extrapolating this concept to e-Commerce suggests that familiarity with an e-Vendor reduces social uncertainty

through an understanding of what is happening, the context, and what will occur next. When a user is asked to enter the credit card information that is necessary to conclude the transaction, the social uncertainty a user may feel about why the e-Vendor wants it and what the e-Vendor will do with the credit card information is, arguably, smaller when the user understands why the e-Vendor needs it. In essence, experience reduces perceived uncertainty about the actions of others. Such familiarity should also increase trust because it is more likely that the e-Vendor will behave as expected, and thus increase trust, when the trusting party is familiar with how the e-Vendor operates and thus have expectations that are grounded in actuality. Previous research supports these conclusions showing that in e-Commerce familiarity and trust are two distinct constructs, with familiarity in a trustworthy company increasing trust in the e-Vendor, and both familiarity and trust increasing online purchase intentions [20].

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